Selamat Siang .

Hallo Good afternoon, how are you today, are you feel so bad ?, Are you Boring?

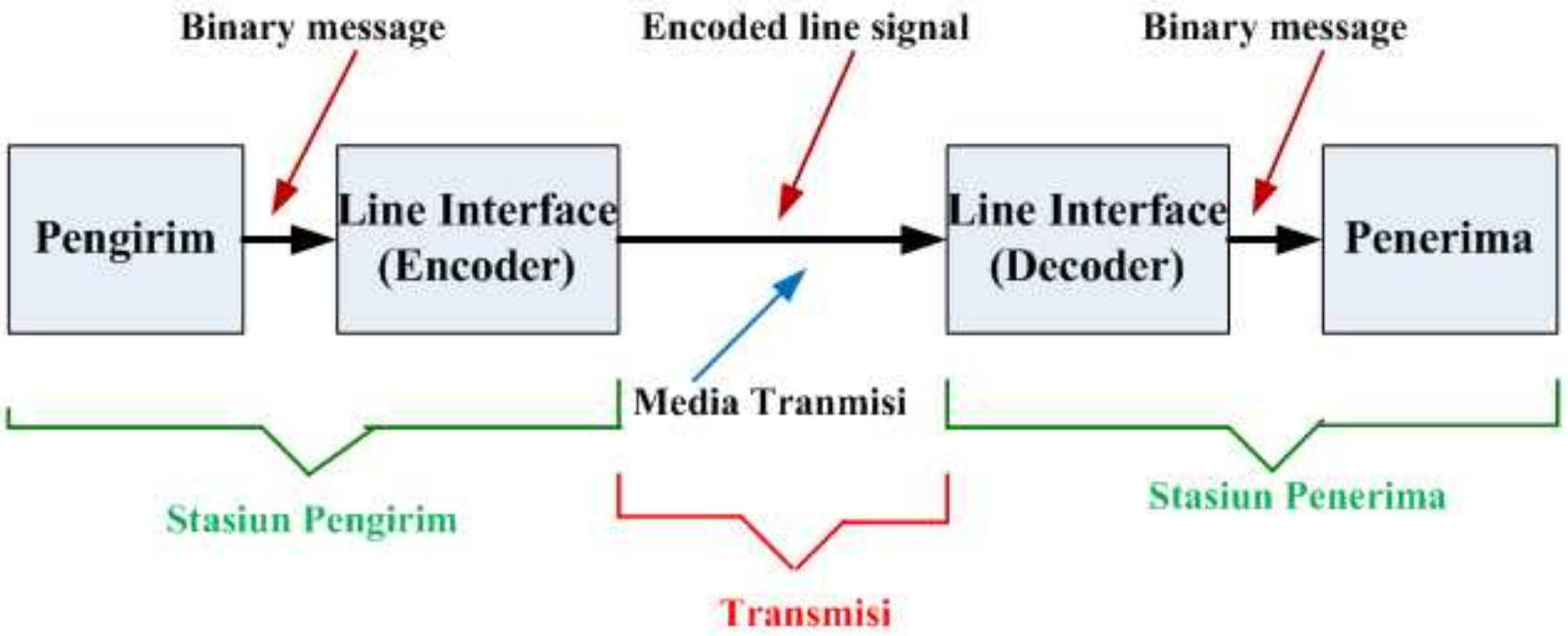
Alhamdulillah.

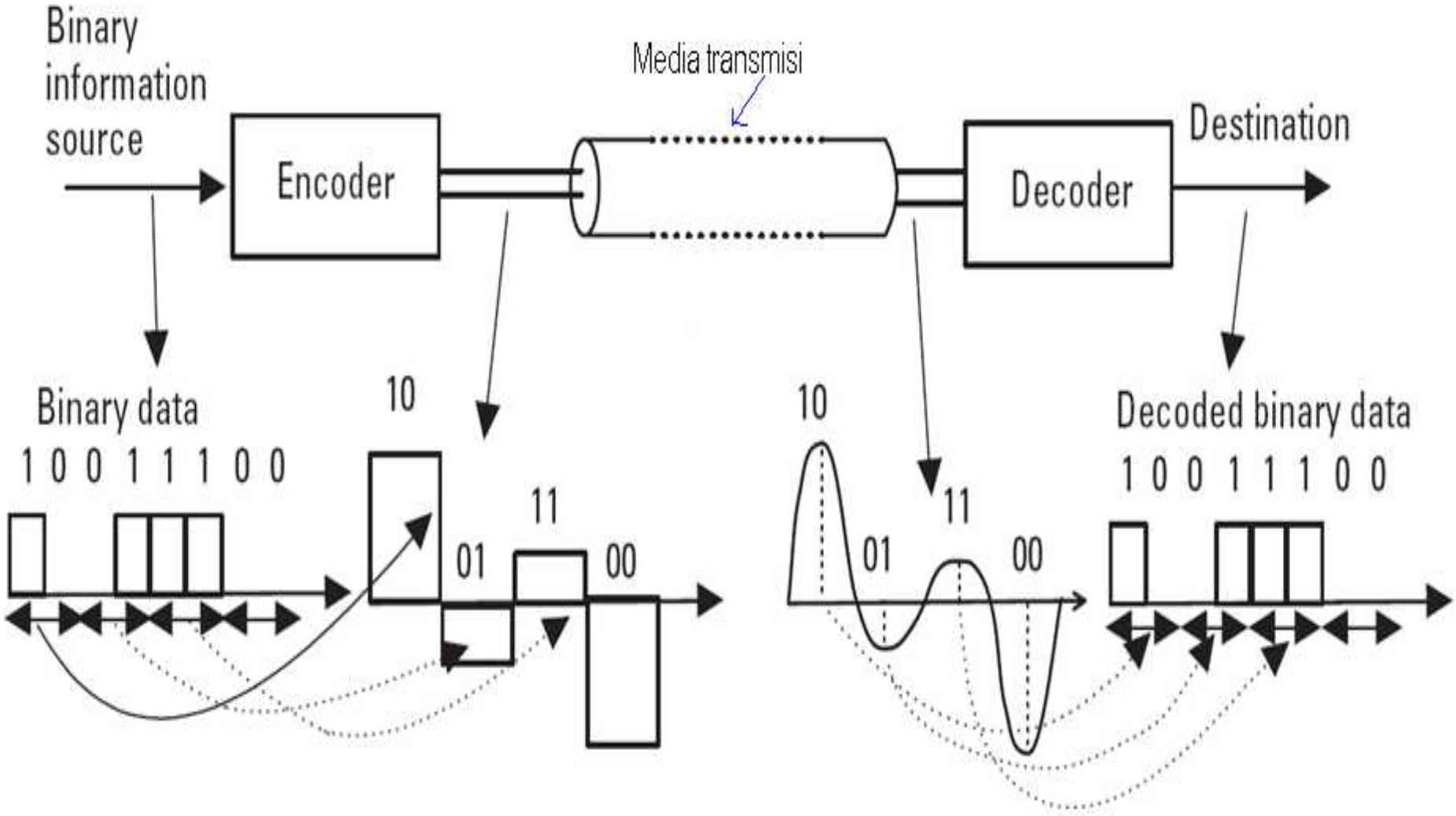
Line Coding
(Data Conditioning)

1. Line Coding

- In order to be transmitted over a digital communications system, an information signal must first be formatted so that it is represented by digital symbols (usually binary digits or bits).
- Next, these digital or binary representations must be converted into electrical waveforms that are transmitted over the communications channel.
- In *baseband* digital transmission, the electrical waveforms (the waveform pattern of voltage or current used to represent the 1s and 0s of a digital data on a transmission link is called *line coding.*
- Sistem yang menggunakan *line coding* tetapi tidak melibatkan modulasi disebut sistem *transmisi baseband*.
  - *Spektrum* hasil pengkodean *tetap* berada di dalam rentang frekuensi pesan asli

- *Line coding* merupakan metoda untuk mengubah simbol dari sumber ke dalam bentuk lain untuk ditransmisikan.
- *Line coding* mengubah pesan-pesan digital ke dalam deretan simbol baru (ini merupakan proses encoding).
- *Decoding* bekerja kebalikannya yaitu mengubah kembali deretan yang sudah dikodekan (encoded sequence) menjadi pesan aslinya.




Binary
information
source
Encoder
Media transmisi
Decoder
Destination
Binary data
1 0 0 1 1 1 0 0
10
01
11
00
10
01
11
00
Decoded binary data
1 0 0 1 1 1 0 0

2.Tujuan Line Coding

- Merekayasa spektrum sinyal digital agar sesuai dengan medium transmisi yang akan digunakan.
- Dapat dimanfaatkan untuk proses sinkronisasi antara pengirim dan penerima (sistem tidak memerlukan jalur terpisah untuk clock).
- Dapat digunakan untuk menghilangkan komponen DC sinyal (sinyal dengan frekuensi 0).
  - *Komponen DC tidak mengandung informasi apapun tetapi menghamburkan daya pancar.*
- Line coding dapat digunakan untuk menaikkan data rate.
- Beberapa teknik line coding dapat digunakan untuk pendeteksian kesalahan.

3. Line Coding Data Biner

- The *binary data* such as the binary 1's and 0's produced by a *PCM encoder* may be represented in various serial-bit *signalling formats* known as line codes.
- There exists many line codes and each has its own particular. advantages and disadvantages depending on the particular application.
- However, there are a number of desirable properties that a line code should exhibit.

a. Signal Spectrum: Several aspects of the signal spectrum are important:

- The spectral occupancy (i.e. the bandwidth) should be as small as possible to ensure good spectral efficiency.
- There should be no dc component as this permits the use of ac coupling via transformer. This provides for electrical isolationand helps reduce the effects of interference.

b. Clock Signal: Synchronization between the transmitter and receiver is of critical importance in digital communications systems.

- Ideally, the spectrum of the line code should contain a frequency component at the clock frequency to permit clock extraction.
- This avoids having to transmit a separate clock signal between the transmitter and receiver.

C. Signal Interference and Noise Immunity:

Ideally, the line code should be rugged in terms of exhibiting an immunity to interference and noise.

- In more technical terms, the line code should have a low probability of error for a given level of transmitted power.
- Certain line codes are more rugged than others, e.g. polar codes have a better error performance compared to unipolar codes.

d. Error Detection: It is useful to have some error detection capability built into the line code to permit transmission errors to be detected more quickly.

e. Transparency: The performance of the line code should be independent of the data, i.e.long strings of binary 1's or 0's should not affect the performance.

f. Cost and Complexity: The line coding scheme should not be excessively complex and/or costly.

4. Line Coding Formats:

The various line coding waveforms can be categorized in terms of the following.

a) The duration of the pulses.

b) The way in which voltage levels are assigned to the pulses.

a). Pulse Duration: There are two classes used here.

- Non return-to-zero (NRZ) where the pulse or symbol duration Ts = the bit period Tb.
- Return-to-zero (RZ) where the pulse or symbol duration Ts < the bit period Tb. Usually Ts = 0.5Tb.
- The pulse duration will usually have an effect on the synchronization properties of the line code (i.e. it determines the presence or absence of a frequency component at the clock frequency).

b). Pulse Voltage Levels: There are many voltage level formats possible:

1) Unipolar
2) Polar
3) Dipolar
4) Bipolar
5) High Density Bipolar substitution (HDBn)
6) Coded Mark Inversion (CMI)

1). Unipolar signalling is where a binary 1 is represented by a high positive level (+A volts) and a binary 0 is represented by a zero level (0 volts), Unipolar menggunakan level +v, 0

- ✓ This is sometimes known as on-off keying (OOK).
- ✓ There are two variations possible:
  - o Unipolar NRZ
  - o Unipolar RZ

❖ **Unipolar NRZ** has the following features:

- Narrow bandwidth
- Significant dc component
- No clock component
- Easy to generate

❖**Unipolar RZ** has the following features:

- Large bandwidth
- Significant dc component
- Clock component present
- More difficult to generate

❖ *In both cases (Unipolar NRZ and Unipolar RZ)there is no error detection capability and the codes*

2). Polar signalling is where a binary 1 is represented by a high positive level (+A volts) and a binary 0 is represented by a negative level (-A volts), Polar (antipodal) menggunakan level +v, -v

- ✓ This is an example of antipodal signalling.
- ✓ There are two variations possible:
  - o Polar NRZ
  - o Polar RZ

❖ **Polar NRZ** has the following features:

- Similar spectrum to unipolar NRZ (narrow bandwidth)
- Significant dc component
- No clock component

❖ **Polar RZ** has the following features:

- Similar spectrum to unipolar RZ (large bandwidth)
- Significant dc component
- No clock component present, but clock extraction possible using rectification.

- ❖ In both cases (Polar NRZ and Polar RZ there is no error detection capability and the codes are not transparent.
- ❖ However, the polar scheme has a better error performance due to its antipodal format.

3). Dipolar coding is designed to produce a spectral null at 0 Hz, i.e. no dc component.

- ✓ The symbol interval Ts is split into half-width pulses.
- ✓ An example of dipolar coding is the Manchester code where a binary 1 is represented by a positive half-width pulse followed by a negative half-width pulse. A binary 0 is represented by a negative half-width pulse followed by a positive half-width pulse.
- ✓ The Manchester code is transparent.
- ✓ Clock extraction is possible at the receiver.
- ✓ Large bandwidth relative to NRZ type coding.
- ✓ Used on IEEE 802.3 Ethernet LANs.
- ✓ Also known as split-phase signalling

4). Bipolar RZ or **Alternate Mark Inversion (AMI)** uses three voltage levels to represent the binary 1's and 0's.

- ✓ A binary 0 is represented by a zero level.
- ✓ A binary 1 is represented by alternating positive and negative pulses (i.e. the alternating mark rule).

✓ This alternating pulse polarity gives bipolar signalling an error detection capability and also produces a spectral null at 0 Hz.

✓ There is no clock component present but clock extraction is possible through rectification.

✓ Bipolar signalling is not transparent, but several techniques have been developed to address this deficiency.

✓ Also known as pseudoternary signalling.

✓ Bipolar (pseudoternary) menggunakan level +v, 0, -v

5). High Density Bipolar substitution (HDBn) is used to counteract the effects of a long strings of binary 0's in the AMI line code.

- ✓ When the number of continuous binary 0's exceeds n they are replaced by a special code sequence.
- ✓ HDB3 is the line code recommended by the ITU-T for PCM systems operating at multiplexed rates of 2, 8, and 34 Mbits/sec. (ITU-T Recommendation G.703). HDB3 is widely used in Europe, include in Indonesia (termasuk di Indonesia).
- ✓ In HDB3, the fourth zero in a string of zeros is "marked", i.e. forcibly set to 1, but in a way that violates the alternating mark rule.
- ✓ Specifically, string of four binary 0's is replaced by either 000V or 100V where V is chosen to violate the alternating mark rule.

- ✓ Furthermore, consecutive violation pulses alternate to avoid introducing a dc component.
- ✓ The HDB3 spectrum is similar to that of the bipolar RZ spectrum.
- ✓ It has a spectral null at 0 Hz and most of the energy is concentrated in a relatively sharp spectral peak around a frequency at half the clock rate.
- ✓HDB3 is well suited to high data rate transmission.

Capek Istirahat sebentar Ya,,,,,,,,.
Boleh kan !!!!!!!
I am so tired, may I take a rest for a moment ?

4. Line Code

❑ Non Return to Zero (NRZ)

- Bit “1” dinyatakan oleh “high signal” selama perioda bit.
- Bit "0" dinyatakan oleh “low signal” selama perioda bit

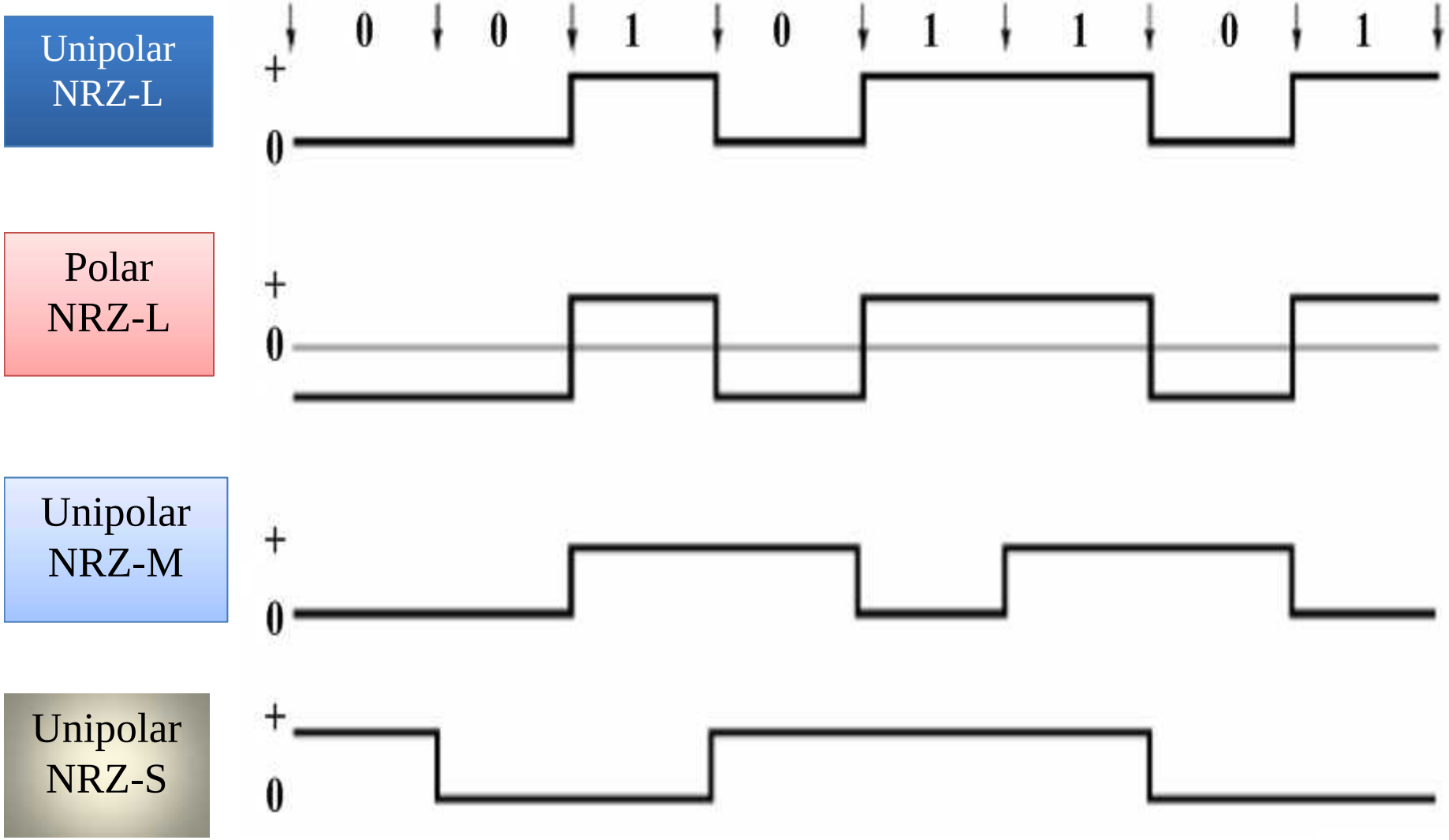
❖ **Kelemahan:**

- Tidak ada informasi timing di dalam bentuk sinyal sehingga sinkronisasi bisa hilang bila muncul deretan 0 yang panjang.
- Spektrum NRZ mengandung komponen DC

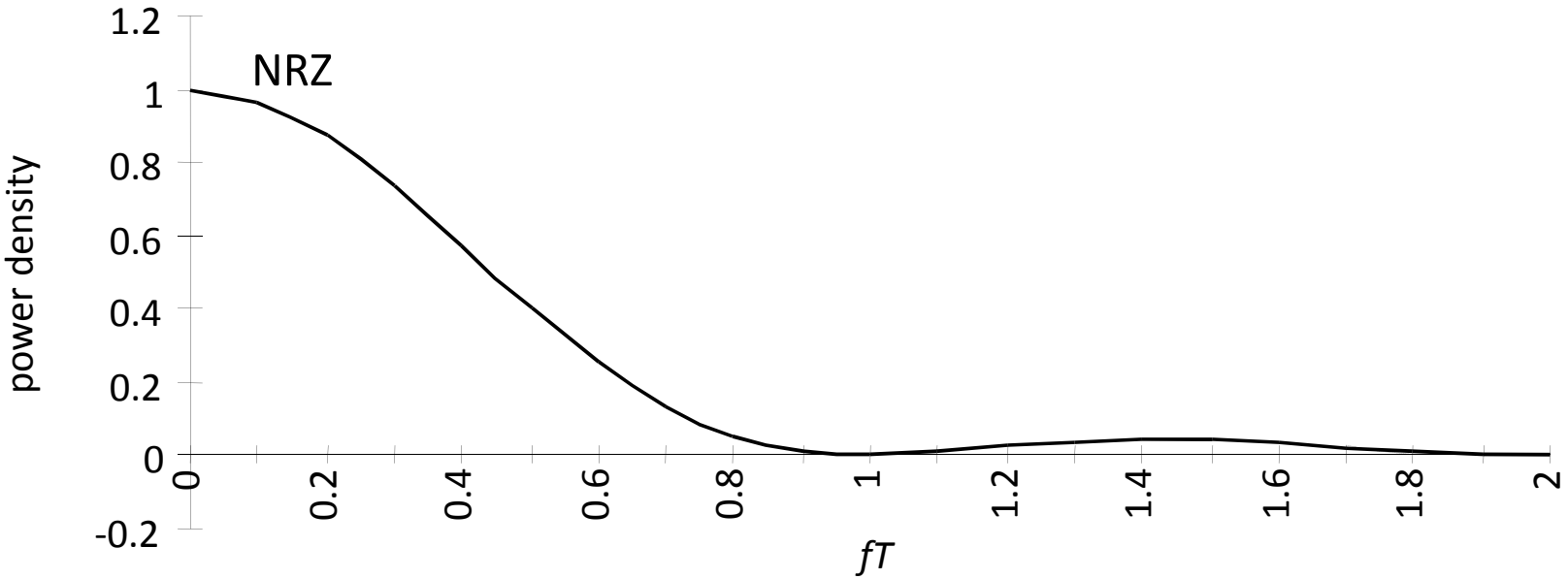
❖ **Varian dari NRZ:**

- **NRZ-L (Non-Return-to-Zero-Level) :** Level konstan selama perioda bit.
- **NRZ-M (Non-Return-to-Zero-Mark):** level berubah bila ada bit “1” .

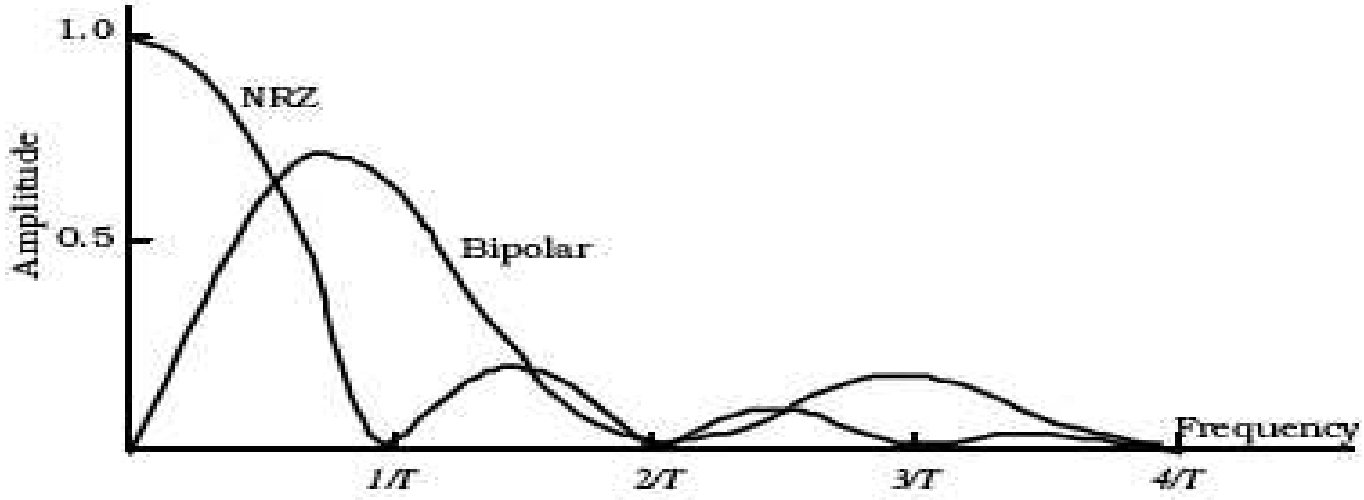
- **NRZ-S (Non-Return-to-Zero-Space):** level berubah bila ada bit “0”
- NRZ bisa unipolar maupun polar.



Spektrum NRZ dan Bipolar



NRZ and Bipolar Spectral Density[3]

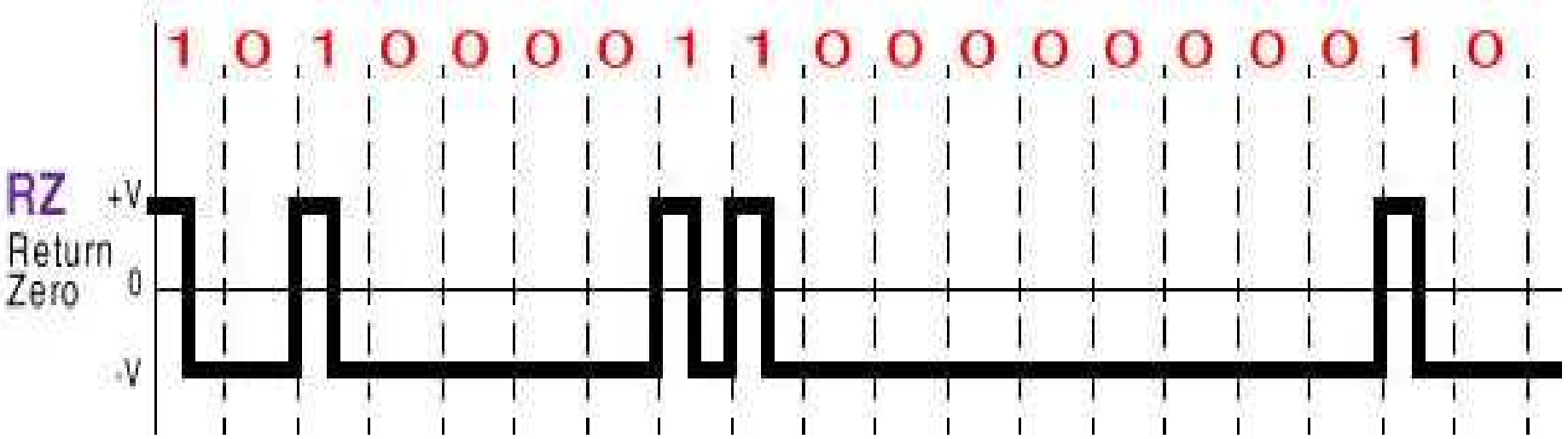


❑ Return to Zero (RZ)

- Bit "1" dinyatakan oleh “high signal” selama setengah perioda bit dan dinyatakan oleh “low signal” pada setengah perioda bit berikutnya.
  - ✓ Memungkinkan pengambilan informasi clock bila ada deretan “1 “ yang panjang.

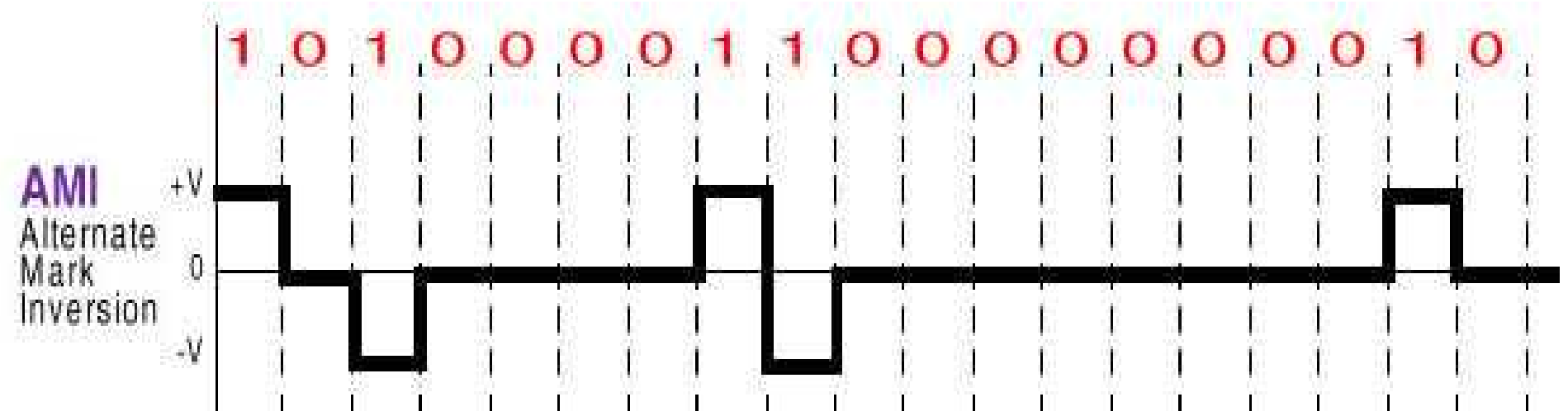
❖ **Kelemahan**

- ✓ Bandwidth yang diperlukan dua kali NRZ
- ✓ Sulit mengambil informasi clock bila ada deretan “0” yang panjang
- ✓ Mengandung komponen DC



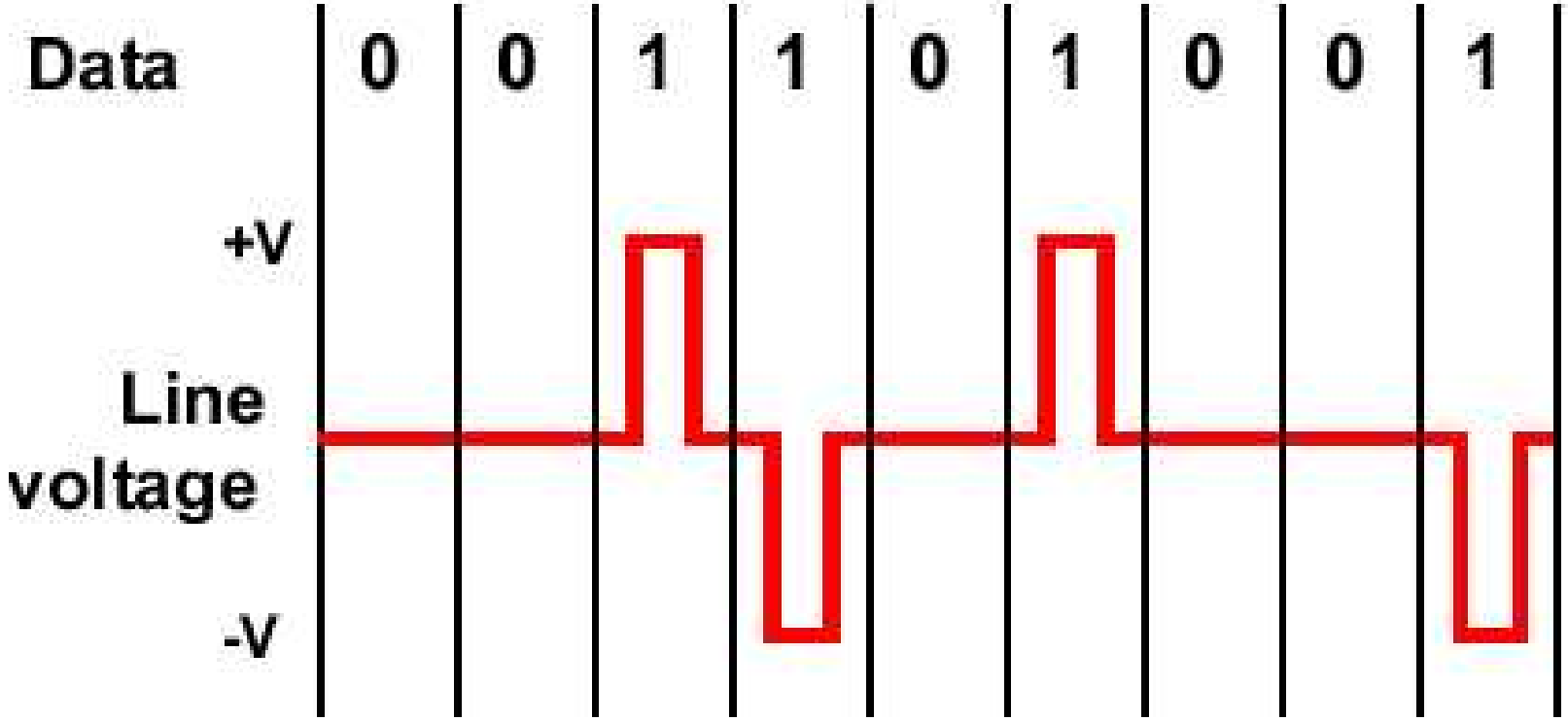
❑ AMI (Alternate Mark Inversion)

- **Pseudoternary code**
  - ✓ Bit "0" dinyatakan sebagai level nol
  - ✓ Bit "1" dinyatakan oleh level positif dan negatif yang bergantian
- **Karakteristik sinyal hasil pengkodean AMI**
  - ✓ Tidak memiliki komponen DC (kelebihan)
  - ✓ Tidak memecahkan masalah kehilangan sinkronisasi bila terdapat deretan “0” yang panjang



AMI

- AMI is usually implemented as RZ pulses, but NRZ and NRZ-I variants exist as well.



❑ HDB3

- Sinyal kode HDB3 Merupakan sinyal bipolar, tiga keadaan dinyatakan sebagai **“B+” , “B-”** , dan **“0”** .
- Berbasis kode AMI
- Sinyal yang **bukan** merupakan data **“0” empat bit berturut-turut,“1”** pada sinyal kode biner dikodekan secara bergantian menjadi **“B+”,** dan **“B-”**pada sinyal HDB3.
- Jumlah bit **“0” berurutan** yang diperbolehkan **maksimum 3**.
- Jumlah bit **“0” berurutan sebanyak 4 bit “0000”**, diganti menjadi **"000V“** atau **"B00V“ .**
  - ✓ "V" (pulsa Violation) adalah pulsa yang menyalahi aturan AMI mengenai perubahan polaritas yang berurutan (“V+” atau “V-”) secara bergantian.
  - ✓ “B" (pulsa bipolar) dengan polaritas sebagai “B+” atau “B-” yang selalu berlawanan dengan polaritas pulsa sinyal HDB3 sebelumnya.

- Aturan penggunaan "000V" atau "B00V" adalah sbb:

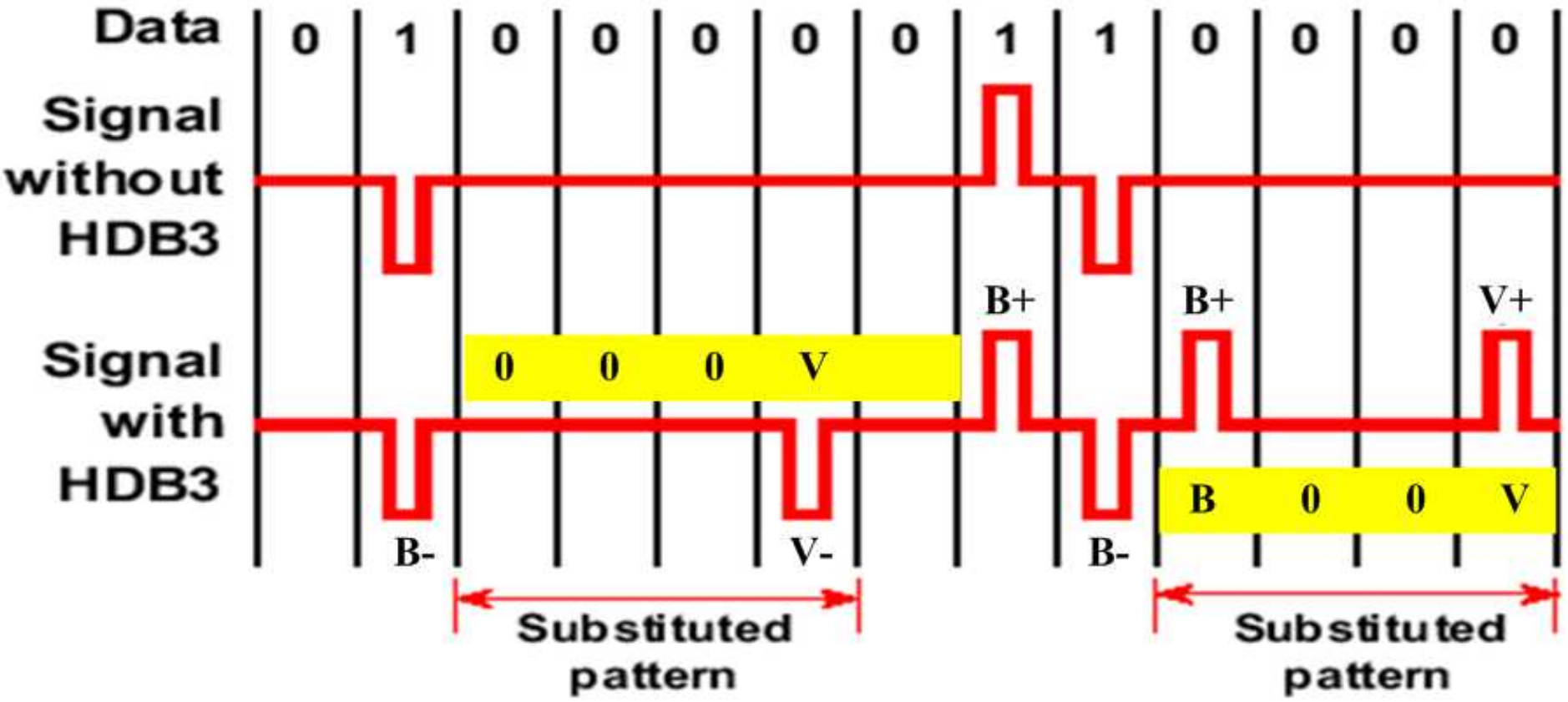
✓ The first “0” bit is coded as a valid pulse (yaitu B + atau B-) according to the AMI rule or a “0 “bit:

- "B00V" If there has been an even number of pulses (of either polarity) since the last “V”, then the first bit is a pulse **(“B+“** atau “**B-**”).
- "000V" If there has been an odd number of pulses (of either polarity) since the last intentional “V”, then the first bit is a “0”.

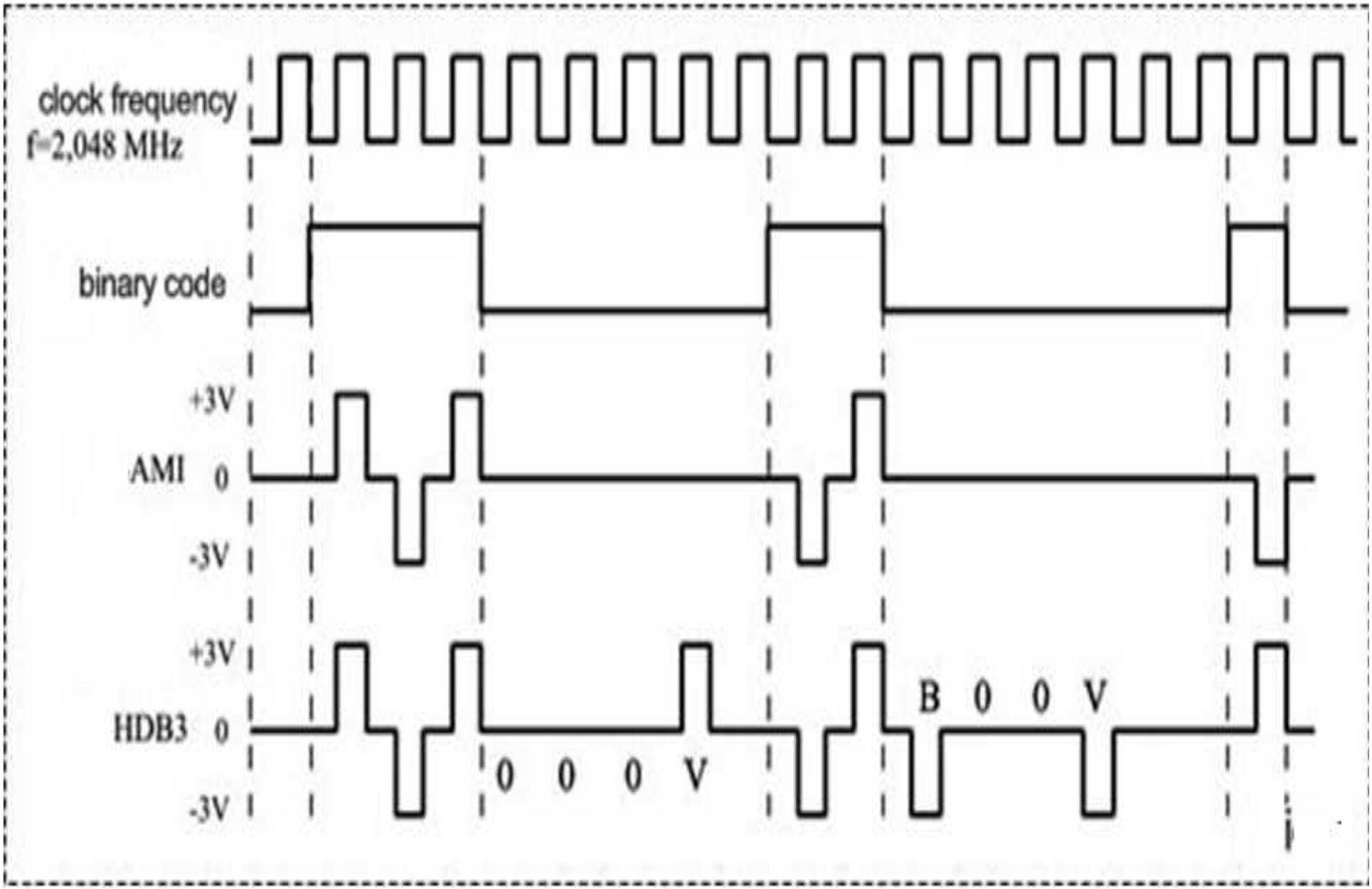
✓ Bit “0” kedua dan bit “0” ketiga pada sinyal biner selalu dikodekan sebagai “0” pada sinyal HDB3.

✓Bit “0” keempat (terakhir) pada sinyal biner, pada sinyal HDB3 selalu dikodekan sebagai “V” dengan polaritas selalu berlawanan dengan polaritas “V” sebelumnya (“V+”, atau “V-”).

High-density bipolar 3 (HDB3)



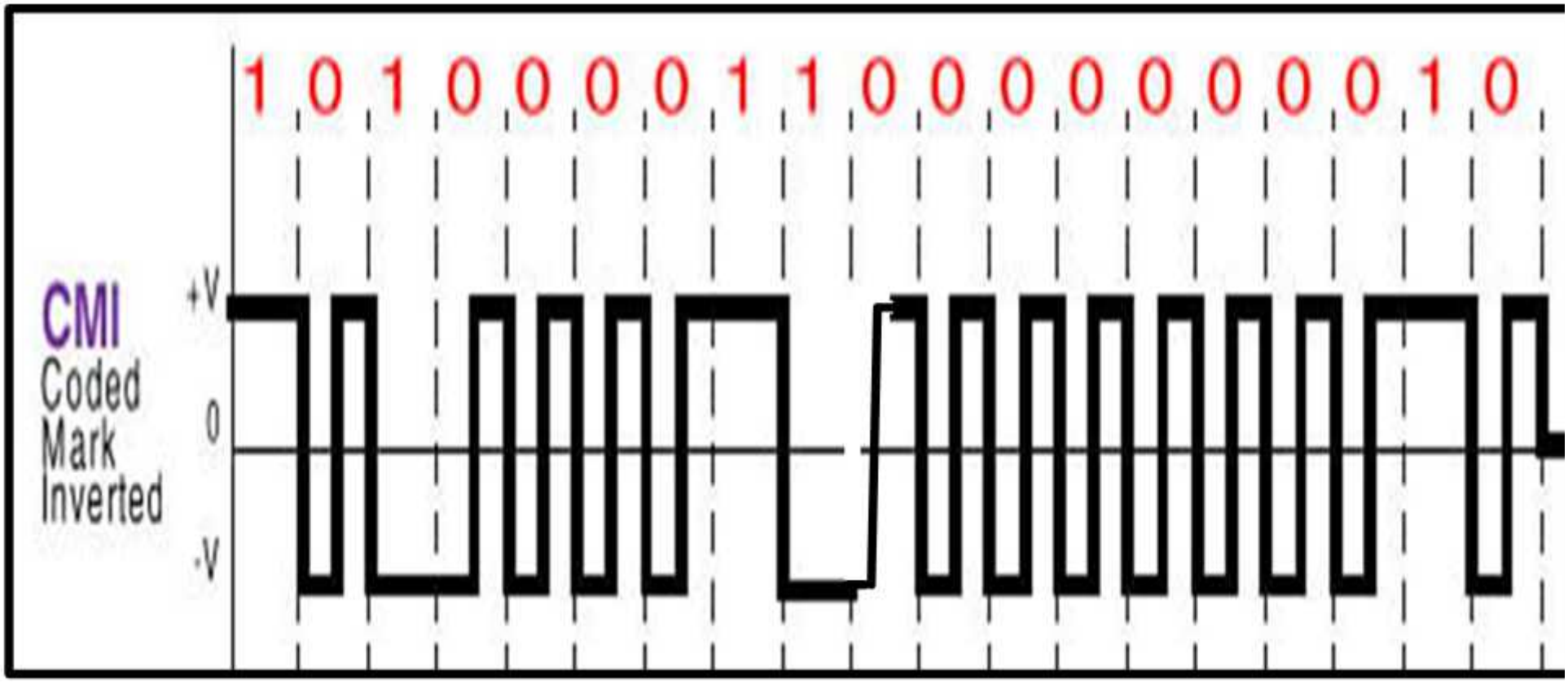
HDB3 dan AMI


clock frequency
f=2,048 MHz
binary code
+3V
AMI 0
-3V
+3V
HDB3 0
-3V
0 0 0 V
B 0 0 V

CMI (Coded Mark Inverted)

- Berbasis AMI
- Digunakan pada transmisis kecepatan tinggi.
- Bit “1” dikirimkan sesuai dengan aturan AMI yaitu bila ada dua “1” berurutan maka pulsa yang menyatakan keduanya harus berbeda polaritas (dikodekan sebagai “+Volt” atau “-Volt” selama satu perioda penuh).
- Bit “0” dinyatakan oleh pulsa dengan setengah perioda pulsa pertama dinyatakan oleh tegangan negatif (-Volt) sedangkan setengah perioda pulsa berikutnya dinyatakan oleh tegangan positif (+Volt).
- Kode CMI memiliki karakteristik berikut:
  - ✓Menghilangkan spektrum sinyal pada frekuensi yang sangat rendah
  - ✓Clock dapat direcovery dengan mudah
  - ✓Bandwidth lebih lebar daripada AMI

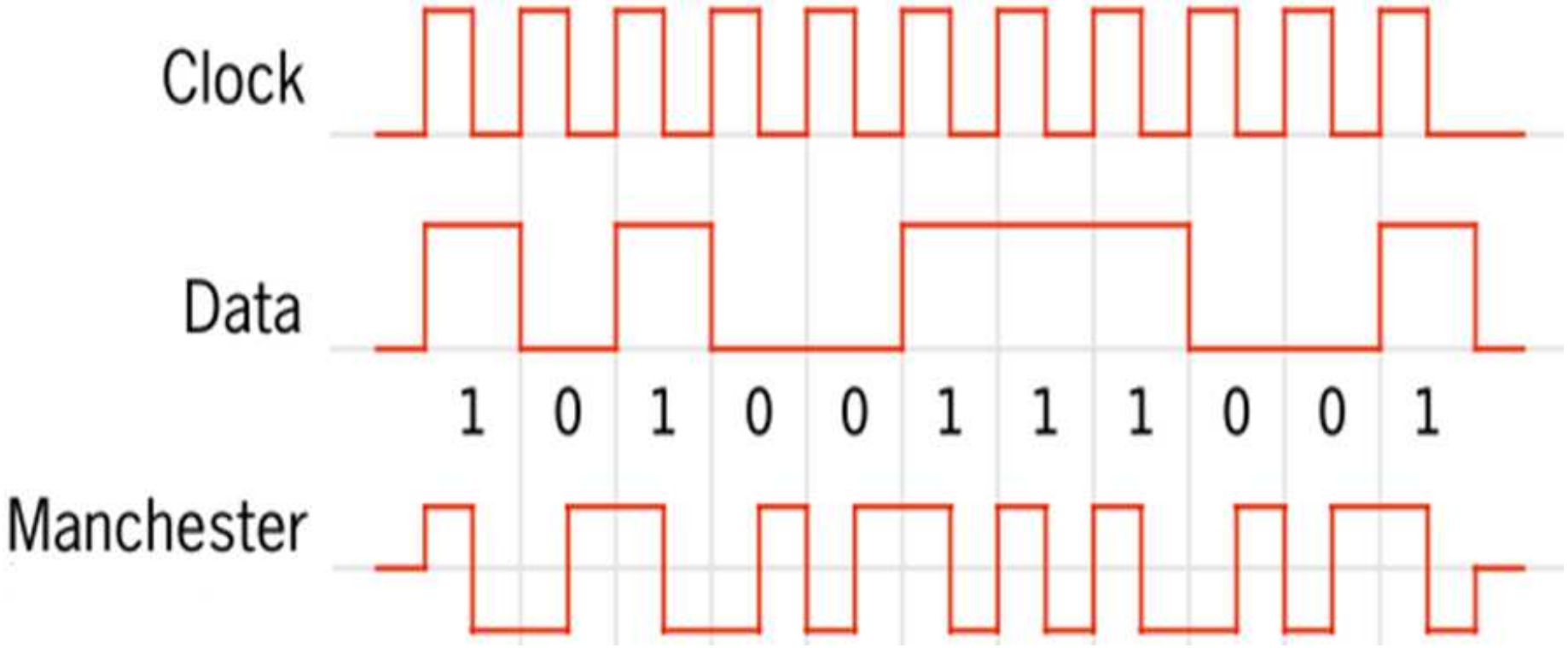
CMI (Coded Mark Inverted)



Manchester

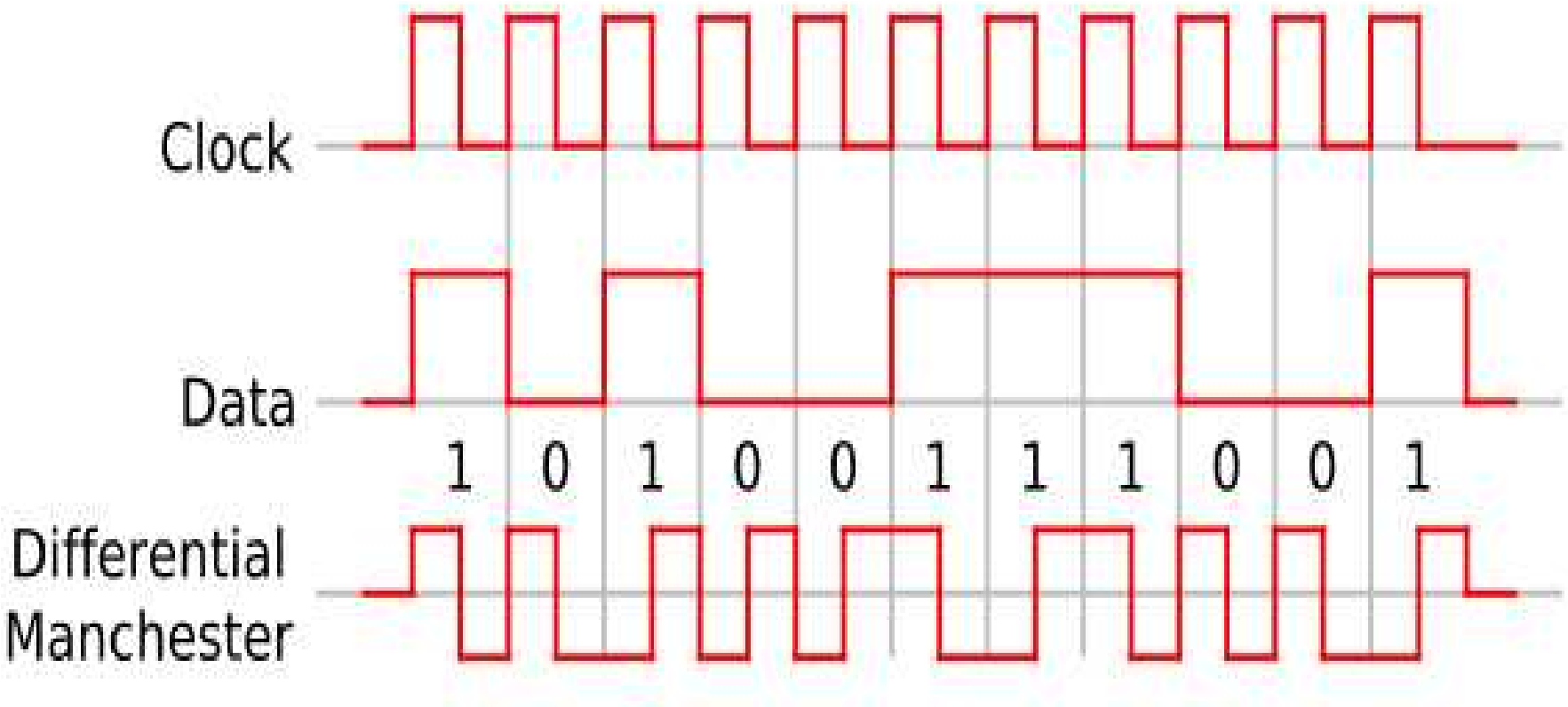
- Bit “1” dinyatakan oleh pulsa yang setengah prioda pertamanya memiliki level high dan setengah perioda sisanya memiliki level low.
- Bit “0” dinyatakan oleh pulsa yang setengah perioda pertamanya memiliki level low dan setengah perioda sisanya memiliki level high.
- Jadi setiap bit dinyatakan oleh pulsa-pulsa yang berganti level pada pertengahan bit.
- Karakteristik Manchester coding:
  - ✓*Timing recovery* mudah
  - ✓Bandwidth lebar

Manchester



Differential Manchester

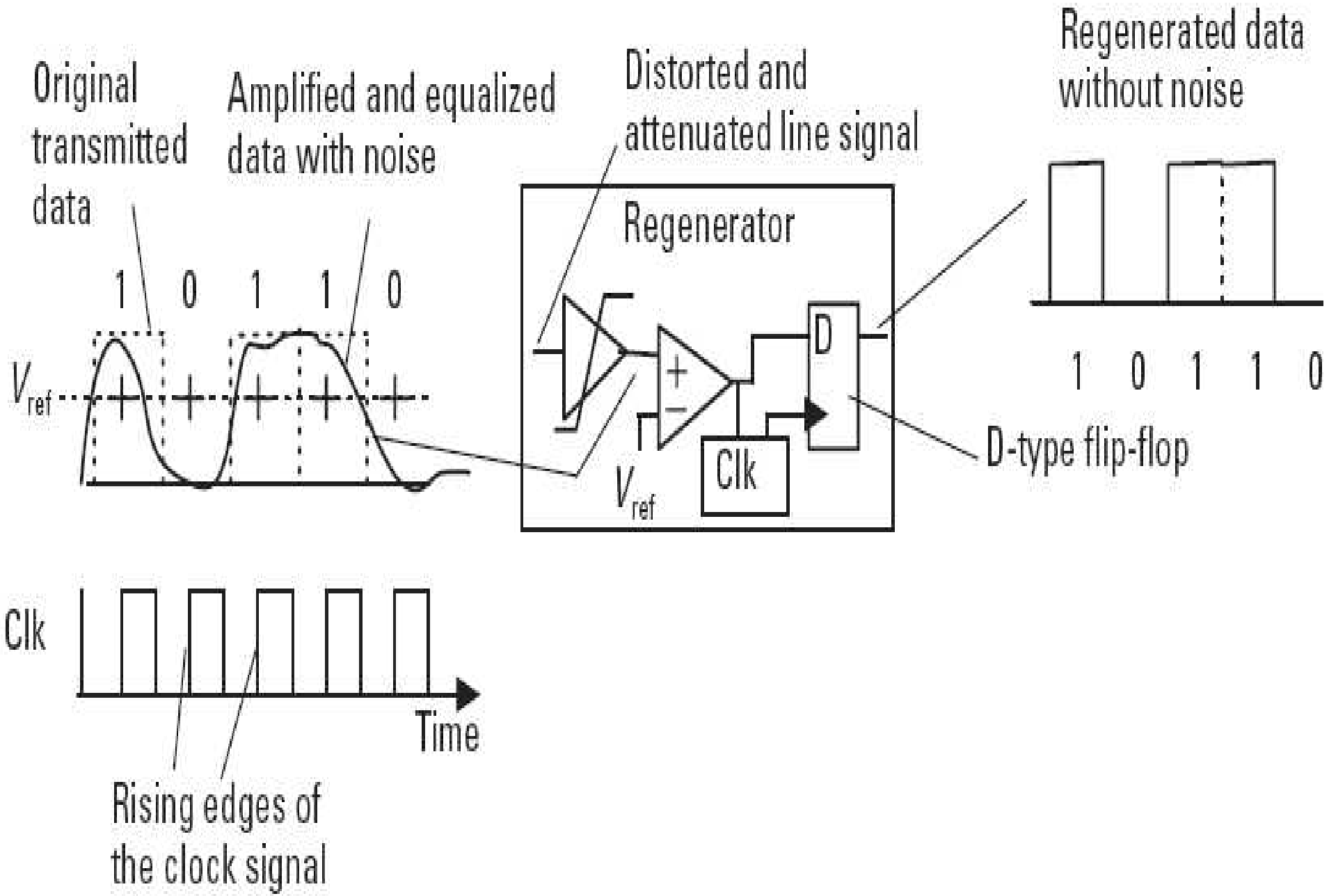
- Setiap bit dinyatakan oleh pulsa-pulsa yang berubah level di pertengahan bit.
- Bit “1” dikodekan dengan tidak adanya transisi level ***di awal bit.***
- Bit “0” dikodekan dengan adanya transisi level *di awal perioda bit*



Regeneration

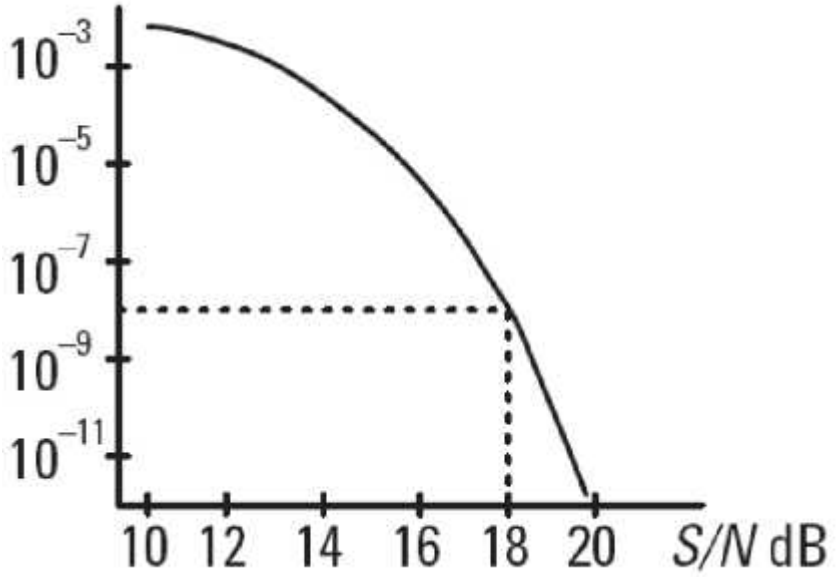
- Pada transmisi jarak jauh, daya sinyal akan teredam sehingga daya yang sampai di penerima bisa jadi sudah sedemikian lemah sehingga tidak dapat dideteksi lagi.
- Pada sistem transmisi analog, digunakan *amplifier/repeater* untuk menguatkan sinyal yang sudah lemah.
- *Amplifier/repeater* selain menguatkan input yang berupa sinyal informasi juga akan menguatkan sinyal noise.
  - ✓Pada penggunaan *amplifier/repeater* yang berulangkali, efek noise akan terakumulasikan sehingga perbandingan Sinyal dengan Noise (S/N) akan semakin mengecil

- Pada sistem transmisi digital, penguatan sinyal dilakukan menggunakan perangkat yang disebut *regenerator* (digital amplifier).
- Suatu *regenerator* terdiri dari *equalizing amplifier*, yg mengkompensasi distorsi dan menapis (mem-filter) *out-of-band noise*, serta sebuah komparator
  - ✓ Keluaran komparator akan *high* jika sinyal input lebih besar daripada V_{ref}, dan akan *low* jika sinyal input lebih rendah daripada V_{ref}
- Sebuah regenator juga mengandung rangkaian pewaktu (timing) yang berfungsi untuk membangkitkan sinyal clock berdasarkan sinyal input yang diterima
- D-flip flop digunakan untuk menentukan apakah sinyal keluaran regenerator high (1) atau low (0) pada saat sinyal clock berada pada kondisi sisi naik (rising edge)
  - ✓ Nilai output akan tetap sampai rising edge berikutnya
- Sinyal hasil regenerasi akan bebas dari noise dan siap ditransmisikan lag


Original transmitted data
Amplified and equalized data with noise
Distorted and attenuated line signal
Regenerated data without noise
Regenerator
1 0 1 1 0
V_{ref}
+
−
D
Clk
V_{ref}
1 0 1 1 0
D-type flip-flop
Clk
Time
Rising edges of the clock signal

- Jika noise terlalu besar, input terhadap komparator bisa jadi berada di atas V_{ref} walaupun sebenarnya sinyal “0” yang sedang dikirimkan
  - ✓Akibatnya akan terjadi kesalahan (error) regenerasi karena yang akan dikeluarkan regenerator adalah sinyal “1” padahal seharusnya adalah sinyal “0”
- Sebaliknya, jika noise terlalu kecil, input terhadap komparator bisa jadi berada di bawah V_{ref} walaupun sebenarnya sinyal “1” yang sedang dikirimkan
  - ✓Akibatnya akan terjadi kesalahan regenerasi karena yang akan dikeluarkan regenerator adalah sinyal nol padahal seharusnya adalah sinyal satu

- Frekuensi error tergantung pada level noise atau d.k.l tergantung S/N
- Jika diasumsikan bahwa noise memiliki distribusi amplituda Gaussian, maka *error rate* (bit error probability) mengikuti kurva *error rate* vs S/N seperti yang terlihat pada gambar
  - ✓Nilai pasti hubungan antara S/N dengan BER berbeda-beda untuk setiap sistem, tetapi bentuk kurva-nya serupa
- Perhatikan bahwa BER akan turun bila S/N semakin tinggi, sebaliknya BER akan naik bila S/N semakin rendah
- Transmisi voice PCM memerlukan syarat BER maksimum 10^{-3}, sedangkan transmisi data memerlukan persyaratan BER yang lebih ketat (maksimum 10^{-9})



Sekian Terima kasih,

Thanks so much, be happy,

don't be sad

Good luck

Minggu depan Quiz